

VILLARRICA, Paraguay

WMO: 862330

Lat: 25.767S

Lon: 56.433W

Elev: 528

StdP: 14.42

Time Zone: -4.00 (PRY)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	41.4	44.9	34.6	29.9	49.6	37.3	33.3	49.9	19.0	69.5	18.2	70.0	2.4	180	0.288

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.5	95.9	75.5	94.3	75.8	92.7	75.8	80.1	89.8	79.4	89.2	78.3	87.9	6.9	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.4	145.5	85.7	76.5	141.0	84.7	75.7	137.0	83.9	44.2	88.9	43.5	90.8	42.4	87.3	85.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.8	15.4	12.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	73.1	81.5	80.1	79.0	73.8	66.4	66.0	65.0	66.9	69.2	74.3	75.8	79.8
	DBStd	9.08	3.20	4.32	4.46	6.24	7.03	8.61	9.34	10.24	8.75	6.16	5.38	4.28
	HDD50	13	0	0	0	0	1	2	5	5	0	0	0	0
	HDD65	468	0	0	0	8	69	97	124	108	55	5	2	0
	CDD50	8449	975	842	900	715	511	482	471	528	575	753	773	924
	CDD65	3428	510	423	435	273	114	127	124	166	180	293	324	459
	CDH74	31787	5530	4293	4222	2120	606	655	754	1394	1665	2656	3053	4839
	CDH80	12961	2497	1877	1845	744	127	141	187	526	650	1057	1203	2107
Wind	WSAvg	4.3	3.2	3.5	3.0	4.0	4.6	4.7	5.1	5.6	5.0	5.1	4.2	3.5
Precipitation	PrecAvg	65.0	6.8	5.4	5.8	6.4	6.0	4.2	3.0	3.4	4.2	7.2	6.7	5.9
	PrecMax	101.3	15.8	17.7	13.5	17.8	17.1	14.2	9.3	11.2	14.1	17.0	19.9	19.6
	PrecMin	42.2	0.2	1.3	0.9	0.0	0.9	0.1	0.0	0.3	0.0	0.1	0.9	1.1
	PrecStd	13.8	3.5	3.4	2.8	4.0	3.9	2.9	2.4	2.7	2.9	3.3	4.0	3.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	97.2	98.2	96.9	92.9	87.7	85.9	87.4	92.2	95.8	95.8	95.2	97.2
		MCWB	74.7	75.8	74.2	76.1	73.7	71.4	69.1	69.3	71.9	77.2	77.3	76.2
	2%	DB	94.9	94.7	94.0	90.7	84.4	83.7	84.7	89.4	91.9	92.7	93.0	94.4
		MCWB	75.2	76.6	75.7	76.1	71.8	71.1	68.3	69.1	71.9	75.8	76.1	76.7
	5%	DB	92.9	92.3	91.6	88.2	81.2	81.2	82.0	86.5	87.7	89.6	90.3	92.4
		MCWB	75.5	76.9	75.8	75.5	70.5	70.3	67.7	68.6	70.4	74.7	75.1	77.0
	10%	DB	91.0	89.8	89.2	85.0	78.0	77.9	78.4	82.7	83.3	86.1	87.3	89.9
		MCWB	76.0	76.6	75.0	74.2	68.9	69.2	67.2	67.5	68.4	72.8	73.9	76.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	80.6	81.4	80.0	79.8	76.0	74.6	73.4	74.6	76.6	79.5	80.1	81.5
		MCDB	90.3	90.2	89.9	89.7	83.5	81.7	81.9	85.4	89.3	92.0	90.4	91.0
	2%	WB	79.6	79.8	78.8	77.7	73.9	72.7	71.3	72.2	74.2	77.6	78.6	79.9
		MCDB	89.1	88.8	88.6	87.2	81.4	80.2	79.2	83.9	86.7	89.4	89.0	89.7
	5%	WB	78.5	78.6	77.7	76.4	72.1	71.1	69.3	70.2	72.0	76.1	77.2	78.7
		MCDB	87.8	87.9	87.0	85.0	78.2	78.6	77.9	81.5	83.9	86.6	86.6	88.5
	10%	WB	77.8	77.7	76.4	75.3	70.2	69.7	67.7	68.4	70.0	74.6	75.8	77.7
		MCDB	86.8	86.3	85.3	83.0	76.0	76.8	76.4	79.0	80.3	83.6	84.4	86.7
Mean Daily Temperature Range	5% DB	MDBR	18.5	17.0	18.3	16.0	15.7	15.2	17.0	18.8	19.2	17.7	18.0	17.6
		MCDBR	21.2	20.5	21.0	19.5	18.2	16.7	19.0	21.1	22.5	21.3	21.4	20.7
		MCWBR	7.2	7.4	7.5	8.1	9.1	8.4	8.7	9.1	10.3	9.4	9.5	8.1
	5% WB	MCDBR	17.4	16.9	17.7	17.3	15.2	15.5	17.6	20.1	19.7	18.3	18.0	17.8
		MCWBR	7.8	7.7	8.0	8.7	9.2	9.0	10.1	11.2	10.2	9.4	9.3	8.4
Clear-Sky Solar Irradiance	taub	0.407	0.410	0.401	0.398	0.374	0.398	0.395	0.498	0.582	0.488	0.417	0.416	
	taud	2.399	2.396	2.404	2.385	2.421	2.347	2.338	2.003	1.799	2.119	2.327	2.358	
	Ebn at Noon	297	292	285	270	263	246	252	235	229	266	292	295	
	Edh at Noon	41	40	37	35	31	32	33	50	68	52	43	42	
All-Sky Solar Radiation	RadAvg	2108	1932	1720	1385	1035	880	1011	1220	1426	1680	2010	2122	
	RadStd	152	152	119	136	142	82	109	83	154	196	149	160	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page